

Instructor Notes — LED & Servo Learn-to-Program Lab (V1)

For instructor use only — not intended for student distribution.

This document accompanies `student-workbook_V3.pdf` and `led_servo_pico_w_lab.ino`. It assumes student laptops have no internet access during class, so any board package or library installation must be done ahead of time.

Required Board Package

- “**Raspberry Pi Pico/RP2040**” by Earle Philhower, installed via Arduino IDE → Tools → Board → Boards Manager.
- Boards Manager URL (needed only once, with internet access, before class): <https://github.com/earlephilhower/arduino-pico>

Required Libraries

- **Servo** — this is bundled with the board package above and is **not** a separate download. Once the board package is installed and “Raspberry Pi Pico W” is selected under Tools → Board, the sketch’s `#include <Servo.h>` resolves automatically with no internet connection required.

No other third-party libraries are required for:

- the base sketch, `led_servo_pico_w_lab.ino`
- any of the 7 challenges in the student workbook’s Section 6 / Appendix A

Every challenge only changes constants, variables, or existing functions already present in the base sketch — none of them add a new `#include`.

Pre-Class Checklist

1. On each laptop, confirm **Raspberry Pi Pico W** appears under Tools → Board → Raspberry Pi RP2040 Boards.
2. Open the base sketch and click Verify (compile) once on each laptop, ahead of class, to confirm the board package and bundled Servo library resolve correctly without needing internet access mid-class.
3. Connect a Pico W to each laptop (hold BOOTSEL, plug in USB) and confirm a port appears under Tools → Port.
4. If you plan to assign **Challenge 4** (5th LED), stock each station with a spare LED, a 330Ω resistor, and a free breadboard slot — the answer key wires the new LED to GP6.

About the Workbook’s Answer Key

Appendix A in the student workbook shows only the lines that change for each challenge, relative to the base sketch, to keep the document short and to discourage copy-pasting a full replacement file without understanding it. If a student needs a fully merged sketch for grading or troubleshooting, combine the shown changes into a copy of `led_servo_pico_w_lab.ino`.